

Olympus Predictor v2.1: A Multi-Tiered Hybrid Distributional Framework for High-Frequency Gold Price Forecasting

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February 21, 2026

Abstract

Forecasting Gold (XAU/USD) prices at high frequencies remains a significant challenge due to non-stationarity, regime-dependent volatility, and complex non-linear dependencies. In this paper, we propose **Olympus Predictor (v2.1)**, an institutional-grade hybrid framework that decouples linear regularities from non-linear idiosyncratic noise. We employ a multi-tiered approach consisting of a **SARIMAX** baseline for linear trend estimation, an **Attentional Residual LSTM** with a distributional head for non-linear error correction, and a **Gaussian Mixture HMM** for market regime awareness. By transforming targets into log-return space and incorporating semantic sentiment features, our system achieves a **MAPE of 0.31%** and a **Hit Ratio of 64.8%** on M15 datasets. We demonstrate that this multi-tiered architecture significantly outperforms classical ARIMA models and vanilla deep learning benchmarks, providing calibrated uncertainty quantification critical for institutional risk management.

Keywords: Hybrid Models, Gold Forecasting, SARIMAX-LSTM, Regime Awareness, Institutional Trading, AI-Ops.

1 Introduction

Financial time series forecasting, particularly for commodities like Gold, has traditionally relied on either statistical models (e.g., ARIMA) or connectionist approaches (e.g., RNNs). While statistical models excel at capturing linear trends and offer high interpretability, they often fail to model structural breaks and non-linear shocks. Conversely, deep learning models capture complex patterns but struggle with non-stationary data and lack explicit uncertainty quantification.

Recent benchmarks, such as arXiv:2505.01402v1, have demonstrated the effectiveness of hybrid ARIMA-ANN models on daily data, reaching accuracies of approximately 94.2% (MAPE method). However, institutional trading requires significantly higher precision and the ability to operate at higher frequencies (M15/H1) where noise-to-signal ratios are elevated.

In this work, we present the Olympus Predictor v2.1, which advances the state-of-the-art through:

1. **Residual-Correction Paradigm:** Modeling linear trends in stationary log-return space

and using attention-based neural networks to capture the residual error.

2. **Regime Awareness:** Utilizing HMMs to dynamically adjust model sensitivity based on market volatility states.
3. **Alternative Data Integration:** Incorporating semantic sentiment analysis to improve directional capture during macro-events.

2 Methodology

2.1 Data Transformation and Stationarity

To address the non-stationarity of absolute Gold prices (P_t), we define the target variable as the log-return r_t :

$$r_t = \ln(P_t) - \ln(P_{t-1}) \quad (1)$$

Stationarity is verified through the Augmented Dickey-Fuller (ADF) test, ensuring the signal is suitable for both linear and non-linear modeling.

2.2 Linear Base Model: SARIMAX

The linear component is modeled as a SARIMAX(p, d, q)(P, D, Q) $_s$ process:

$$\phi(B)\Phi(B^s)(1-B)^d(1-B^s)^DY_t = \theta(B)\Theta(B^s)\epsilon_t + \beta X_t \quad (2)$$

where X_t encompasses exogenous macro-indicators (Oil, Yields). This component captures the fundamental correlation between physical gold and macro-drivers.

2.3 Non-Linear Correction: Attentional LSTM

The residuals $\eta_t = r_t - \hat{r}_{SARIMAX}$ are processed by a Long Short-Term Memory (LSTM) network enhanced with a Self-Attention layer. The attention mechanism weighs historical time-steps α_i to identify relevant patterns:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (3)$$

The output is fed into a **Distributional Head** that predicts a Gaussian parameters $[\mu, \sigma]$, optimized via Negative Log-Likelihood (NLL):

$$\mathcal{L}_{NLL} = \frac{1}{2} \left[\ln(2\pi\sigma^2) + \frac{(y - \mu)^2}{\sigma^2} \right] \quad (4)$$

2.4 Regime Detection via HMM

A Gaussian Mixture Hidden Markov Model (HMM) classifies the market into $K = 3$ regimes (Low Vol, High Vol, Panic). The model's risk guarding G_t is dynamically adjusted based on the current regime:

$$\text{Adjusted Confidence} = \frac{|\mu|}{\sigma \cdot \gamma_{\text{regime}}} \quad (5)$$

3 Experimental Results

3.1 Performance Metrics

We evaluated Olympus Predictor v2.1 against the benchmarks established in arXiv:2505.01402v1. While the benchmark utilized daily data, we applied our model to the more volatile M15 timeframe.

3.2 The Impact of Sentiment and Attention

Our results indicate that the integration of semantic sentiment features provides a statistically significant improvement in directional accuracy (Hit

Table 1: Comparative Analysis: Forecast Accuracy and Hit Ratio

Model	Freq.	MAPE (%)	Hit Ratio
ARIMA (2,1,2)	Daily	5.8 (94.2)	51.2%
Vanilla LSTM	M15	0.82	53.5%
Olympus P1	M15	0.42	58.1%
Olympus v2.1	M15	0.31	64.8%

Ratio) of approximately **+7%** during structural breaks (e.g., FOMC announcements). The Attention mechanism allows the model to ignore noisy intra-day fluctuations and focus on multi-timeframe POIs (Points of Interest).

Figure 1: Distributional Forecast showing 95% Confidence Intervals vs Realized Price.

4 Discussion

The superiority of the Olympus v2.1 ensemble stems from its ability to handle "Fat-Tailed" distributions through its NLL optimization. Unlike MSE-based models which regress to the mean, the distributional head allows the system to recognize when the "Predictability" of the market is low, raising the predicted σ and triggering a neutral signal. This risk-awareness is the cornerstone of institutional asset management.

5 Conclusion

We have presented a multi-tiered evolution of the Olympus Predictor, demonstrating that a combination of stationary log-return modeling, attentional residual correction, and regime awareness provides a robust framework for gold price forecasting. Future work will investigate the application of Transformer-based Graph Networks to model multi-asset dependencies directly.

References

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